

1

Conditioning c

FIMr1p1 ensemble guidance

PR + T2M weekly summaries:
mean, spread, q10, q90, count

Observed predictors

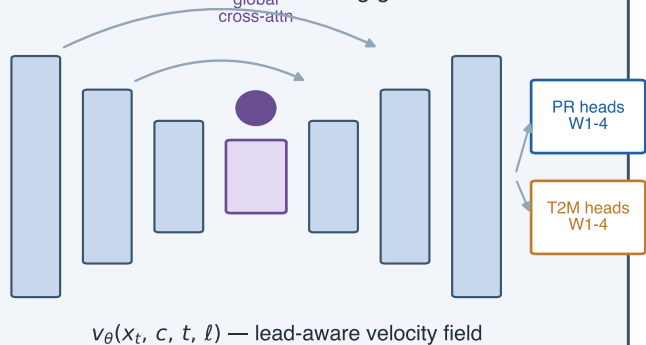
SST, SSS, soil moisture, IVT,
Z500*, U250, MJO (4 prior weeks)

Static + calendar

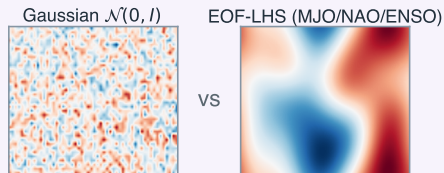
elevation, land mask, lat/lon
encodings, season, lead week

2

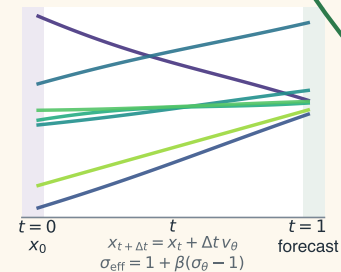
Conditional flow-matching generator

velocity v_{θ}

3

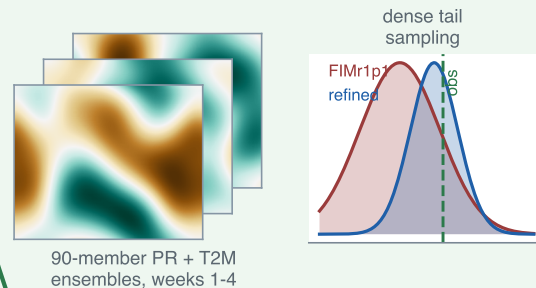
Physically informed stochastic prior x_0 

ODE sampling (50 Euler steps)



4

Weekly probabilistic forecasts



Verification

CRPS

RMSE / bias
/ corr

BSS

spread skill

season x lead

event tails
(q95/q99)